

TRAN NGUYEN DUY BAO

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PROFESSIONAL SUMMARY

AI Engineer with ~3 years of production experience building and optimizing LLM systems, NLP pipelines, and multimodal AI applications in healthcare and media. Specializes in LLM inference optimization on H100 GPU clusters, multi-agent orchestration, and RAG architectures. Combines rigorous MLOps practices with clean software engineering to ship reliable, enterprise-grade AI systems.

TECHNICAL SKILLS

Core: Large Language Models, Fine-Tuning, RAG, Prompt Engineering, Multi-Agent Systems

Frameworks: vLLM, PyTorch, Hugging Face Transformers, LangChain, LangGraph, LangFuse, Camel-AI

Data & ML: Dagster, PySpark, Kafka, RabbitMQ, XGBoost, Scikit-learn, SpaCy, YOLOv8

Backend & DBs: FastAPI, Flask, PostgreSQL, MongoDB, Pinecone, Elasticsearch

Infrastructure: Docker, GCP, AWS, NVIDIA H100, Git

Languages: Python, SQL, JavaScript

PROFESSIONAL EXPERIENCE

Trivita AI

AI Engineer

Ho Chi Minh City, Vietnam

Oct. 2025 – Present

- Optimized inference throughput for **MoE LLMs up to 397B parameters** (Qwen3, MedGemma) on H100 clusters via tensor/expert parallelism, KV cache tuning, and quantization with vLLM.
- Built a two-tier **LLM-as-a-Judge evaluation system** combining heuristic pre-filtering with Prometheus evaluators to score medical dialogues on hallucination rate, clinical accuracy, and treatment compliance.
- Fine-tuned **PhoBERT** for Vietnamese ICD-10 code classification on a heavily imbalanced long-tail dataset; applied augmentation and resampling to recover minority class recall.
- Monitored latency, contextual accuracy, and prompt efficiency via **LangFuse**, driving prompt refinements that measurably reduced token costs.

MedVita – Synthetic Medical Conversation Generation Pipeline

2025 – Present

- Designed a **Dagster orchestration pipeline** automating end-to-end generation and evaluation of synthetic doctor-patient conversations using a strict 4-stage clean architecture: preprocessing, inference, postprocessing, and evaluation.
- Architected multi-agent systems with **LangChain and Camel-AI** to simulate realistic clinical dialogues across 10+ specialized medical departments, with dynamic patient persona generation and scenario synthesis.

MCV Complex

AI Engineer

Ho Chi Minh City, Vietnam

Jul. 2023 – Oct. 2025

- Shipped 4 production AI systems across healthcare and media verticals, owning full-stack delivery from model selection through deployment on GCP and AWS.
- Introduced a modular inference engine toggling between local open-source models (vLLM, Ollama) and cloud APIs, cutting inference costs for standard-tier users.

SpaceOne – Real-Time Media Summarization & Sentiment Analysis

2023 – 2025

- Designed a two-module pipeline: isolated vocals with **Spleeter**, transcribed with **Whisper**, then summarized via structured **GPT-4** prompts over LangChain; integrated into a **Kafka** streaming architecture for real-time processing at scale.
- Fine-tuned **PhoBERT** on domain-specific Vietnamese social media comments for 3-class sentiment classification, reaching **F1-score of 0.84** on real-world data.

Doctor Assistant – Multimodal Medical Chatbot

2023 – 2025

- Built a **LangGraph**-powered conversational agent with memory and tool use, enabling multi-turn clinical dialogue across text, PDF, speech (Whisper), and image (PIL) inputs.

- Embedded structured medical literature into **Pinecone** vector DB for RAG-powered responses; designed JSON-constrained prompt templates to enforce structured LLM output.

Spooface – FaceID with Anti-Spoofing Detection 2024

- Fine-tuned **YOLOv8** to classify real vs. spoofed inputs (printed images, screens, masks) on a custom dataset with synthetic and real-world scenarios; integrated **DeepFace** for post-liveness identity verification.
- Achieved **92% spoof detection accuracy** with sub-1s latency in live event conditions, reducing unauthorized check-ins by over **85%**.

Diabetes Risk Prediction from Tabular Survey Data 2025

- Trained 10+ regression models on OCR-parsed survey data with log-transformed glucose targets, ADASYN oversampling, and RobustScaler normalization; **SVR achieved $R^2 > 0.85$** with ~5% mean prediction error.

EDUCATION

Ton Duc Thang University
B.Sc., Computer Science

Ho Chi Minh City, Vietnam
Graduated Jun. 2025

GPA: 8.01 / 10.0 | Top 10, Recursion Hackathon 2022 | English Proficiency: C1

- Capstone: Fine-tuned **YOLOv8** on NIH ChestX-ray14 with CLAHE and elastic distortion augmentation for radiology abnormality detection; achieved **70.2% mAP** and 60.4% recall on held-out test set.

REFERENCES

Available upon request:
Dr. Nguyen Trang Thao | Former PhD at TDTU Lab | thao.nguyentrang@vlu.edu.vn
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Mr. Ong Xuan Hong | Head of Data | Trivita AI | ongxuanhong@gmail.com